

AIR QUALITY ANALYSIS

India Air Quality Analysis Report

AUTHOR

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DATASET

India Air Quality Data (synthetic, 14 cities, Jan 2022–Dec 2023)

ANALYSIS ENVIRONMENT

Microsoft Excel Online (Microsoft 365)

Executive Summary

This report covers an air quality analysis of 10,220 daily observations across 14 Indian cities between January 2022 and December 2023. The dataset was checked for completeness, extended with seven engineered features to support time-based and categorical analysis, explored through eleven PivotTable views, and summarized in a single-page Excel dashboard built around four KPI cards and five supporting charts. Delhi recorded the highest average AQI (247.9) and Bengaluru the lowest (68.4); winter months carried the heaviest pollution load, and AQI showed a moderate negative relationship with both temperature ($r = -0.58$) and humidity ($r = -0.48$). Missing values across the dataset stayed under 1% per column and were retained rather than imputed, since the sample size and overall completeness meant the gaps carry little weight in any aggregate calculation (see Section 5 and the Decision Framework below).

01 — Project Overview

This project looks at air quality data collected from 14 Indian cities during 2022–2023. The goal was to check the dataset's quality and completeness, engineer features that support time-based and categorical analysis, explore pollution patterns across cities, seasons, and time, identify the most and least polluted cities, and present the findings through a clear Excel dashboard.

The full workflow — from baseline data quality assessment through feature engineering, exploratory analysis, and dashboard design — was carried out in Microsoft Excel Online, using native Excel Tables, formulas, PivotTables, and PivotCharts rather than an external BI tool or scripting language. This report walks through each stage of that workflow along with the exact formula used at every step, so the analysis can be reproduced or reviewed by anyone working from the same source file.

RATIONALE

Air quality data feeds into public-health advisories, pollution-control policy, and city-level environmental planning. An analysis is only as reliable as the completeness and cleanliness of the data behind it, which is why this project treats data quality assessment as its own documented stage rather than a quick first look at the file.

02 — What Is a Data Quality Assessment?

A data quality assessment is the process of checking how complete, consistent, and reliable a dataset is before drawing any conclusions from it. Rather than assuming a dataset is ready for analysis just because it loads without error, this stage measures the dataset against concrete criteria — row counts, missing-value rates, minimum and maximum bounds, data types, and the number of distinct values per column — so that any downstream aggregate or chart carries a known margin of uncertainty.

This project follows a five-stage workflow, each stage labeled below with its corresponding industry term:

- **Data Profiling / Baseline Assessment** — establish dataset structure and characteristics prior to any analysis

- **Data Quality Assessment** — measure missing values, ranges, data types, and completeness per column
- **Outlier Assessment** — identify statistically extreme values and confirm whether they are valid
- **Feature Engineering** — derive new columns that simplify time-based and categorical analysis
- **Exploratory Data Analysis & Dashboarding** — surface patterns via PivotTables and present them visually

DECISION FRAMEWORK: RETAIN vs. IMPUTE vs. DROP

Each remediation choice in this project follows one of three treatments, stated once here and applied consistently throughout:

Retain as-is — used when missing values fall under 1% of a column and removing or estimating them would introduce more distortion than the gap itself. This was the treatment applied to every numeric column in this dataset (see Section 5).

Drop the row — reserved for exact full-row duplicates only. No duplicate rows were found in this dataset, so this treatment wasn't needed.

Flag / leave as a documented limitation — used when a platform constraint prevents a fix within the current environment, as with certain Excel Online dashboard limitations noted in Section 12.

03 — Tools and Environment

Metric	Result
Microsoft Excel Online (Microsoft 365)	Browser-based spreadsheet tool used for the entire workflow — profiling, feature engineering, PivotTables, and dashboard.
Excel Tables	Used to structure the raw and modified datasets for structured-reference formulas ([@Column]).
PivotTables & PivotCharts	Used for all exploratory analysis and most of the dashboard visuals.
Excel Functions	COUNTBLANK, MIN, MAX, YEAR, MONTH, TEXT, IF, OR, ROUNDUP, WEEKDAY, COUNTA/UNIQUE.

04 — Dataset Overview & Baseline

INDUSTRY STAGE: DATA PROFILING / BASELINE ASSESSMENT

Before any modification, the raw dataset was inspected to confirm its structure. The table comprises 11 original columns describing daily air quality readings per city: Date, City, AQI, PM2.5, PM10, NO2, SO2, CO, O3, Temperature, and Humidity, spanning 10,220 records collected from 14 cities between January 1, 2022 and December 31, 2023.

FORMULA

```
=COUNTA(UNIQUE(City))
```

What this does: UNIQUE() returns the distinct list of city names in the dataset, and wrapping it in COUNTA() converts that list into a single count. This confirmed that the dataset spans exactly 14 cities, a figure used as a control total for every city-level breakdown that follows.

Metric	Result
Total records	10,220
Cities covered	14
Date range	Jan 1, 2022 – Dec 31, 2023
Original columns	11 (Date, City, AQI, PM2.5, PM10, NO2, SO2, CO, O3, Temperature, Humidity)
Columns after feature engineering	18

05 — Stage 1: Data Quality Assessment

INDUSTRY STAGE: HANDLING MISSING DATA

Every column was checked for missing values, its observed minimum and maximum, its data type, and — for City — its count of distinct values. This column-by-column approach keeps coverage consistent regardless of where a gap happens to fall.

FORMULA

=COUNTBLANK('Original Dataset'!D2:D10221)

=Missing Values/10220 → formatted as a percentage

What this does: COUNTBLANK() counts empty cells in a column across all 10,220 rows in a single pass. Dividing that count by the fixed row total and formatting the result as a percentage produces a comparable completeness score for every column, shown in the table below.

Column	Missing Values	% Missing	Min	Max	Data Type
Date	0	0.00%	01-Jan-2022	31-Dec-2023	Date
City	0	0.00%	—	—	Text (14 unique)
AQI	0	0.00%	25	508	Number
PM2.5	100	0.98%	11.8	283.6	Number
PM10	77	0.75%	18.9	560	Number
NO2	89	0.87%	7.1	123.1	Number
SO2	95	0.93%	3.3	39.6	Number
CO	97	0.95%	0.28	4.65	Number
O3	74	0.72%	10.8	98.6	Number
Temperature	87	0.85%	9.1	42.8	Number
Humidity	78	0.76%	10	98	Number

DECISION RATIONALE

Every column's missing-value rate falls under 1%. At this scale, dropping rows would discard otherwise-valid readings across the other 10 columns of each record, and imputing values (for example, filling PM2.5 gaps with a city-level average) would introduce estimated numbers into a pollutant dataset without a clear basis for doing so — unlike, say, deriving a known state from a known city. Missing values were therefore kept as true blanks rather than dropped or estimated, keeping the dataset close to its original form while staying transparent about where the gaps sit.

06 — Stage 2: Duplicate Record Check

INDUSTRY STAGE: DEDUPLICATION

A duplicate record, in this context, is one where every column holds a value identical to another row. The dataset was reviewed for exact full-row repeats across all original columns before any feature engineering was applied, since duplicate rows would otherwise inflate every downstream city-level and time-based average.

What this does: The dataset was checked for rows matching on every original column simultaneously. No duplicate rows were found, so no rows were removed at this stage — the 10,220-row baseline established in Section 4 carried through unchanged into feature engineering and analysis.

07 — Stage 3: Outlier Assessment

INDUSTRY STAGE: OUTLIER DETECTION & TREATMENT

Extreme values were reviewed across all pollutant and weather columns before analysis, using the minimum and maximum bounds established in Section 5 as a starting point. Unlike a rule-based sentinel check on a transactional dataset, air pollution and weather readings can genuinely swing to extreme values during real events — a severe smog episode or an unusually hot day isn't a data-entry error, it's the data.

DECISION RATIONALE

No impossible values were found (no negative pollutant concentrations, no AQI values outside a plausible range, no temperature or humidity readings outside a realistic band for the Indian climate). Because air pollution can naturally produce extreme spikes — and because the maximum AQI value of 508 corresponds to a real, if rare, "Severe+" pollution event rather than a repeated sentinel value — the observed high values were kept rather than removed. This follows the same investigate-before-removing principle used for sentinel detection in transactional datasets, applied here to a domain where extreme values are expected rather than anomalous.

08 — Stage 4: Feature Engineering

INDUSTRY STAGE: FEATURE ENGINEERING

Seven new columns were derived from the two original fields (Date and AQI) to simplify PivotTable analysis and enable seasonal, temporal, and categorical breakdowns that the raw dataset couldn't support on its own.

8.1 — Year, Month, and Month Name

FORMULA

```
Year = YEAR([@Date])
Month = MONTH([@Date])
Month Name = TEXT([@Date], "mmm")
```

What this does: YEAR() and MONTH() pull numeric components directly from the Date column, which helps with filtering, grouping, and dashboard performance compared to filtering on the full date. Month Name was derived afterward using TEXT() rather than typed manually, so month names stay tied to Month Number and continue sorting in chronological order inside PivotTables and charts, rather than defaulting to alphabetical order.

8.2 — Quarter

FORMULA

```
Quarter = "Q"&ROUNDUP([@Month Number]/3,0)
```

What this does: Dividing the month number by 3 and rounding up maps months 1–3 to Q1, 4–6 to Q2, and so on, producing a label that supports quarter-level analysis — a grouping often used in place of month-by-month comparison.

8.3 — Season

FORMULA

```
Season = IF(OR([@Month]=12,[@Month]<=2),"Winter",
  IF([@Month]<=5,"Summer",
    IF([@Month]<=9,"Monsoon","Post-Monsoon")))
```

What this does: A nested IF maps each month number to one of four Indian seasonal bands — December through February as Winter, March through May as Summer, June through September as Monsoon, and October–November as Post-Monsoon — supporting the season-level pollution comparison in Section 9.

8.4 — AQI Category

FORMULA

```
AQI Category = IF([@AQI]<=50,"Good",
  IF([@AQI]<=100,"Satisfactory",
    IF([@AQI]<=200,"Moderate",
      IF([@AQI]<=300,"Poor",
        IF([@AQI]<=400,"Very Poor",
          IF([@AQI]<=450,"Severe","Severe+"))))))))
```

What this does: Raw AQI numbers aren't intuitive on their own, so this formula translates each reading into the standard national AQI category using the same breakpoints used for public air-quality advisories. This turns a

continuous number into a category that both PivotTables and non-technical dashboard viewers can read at a glance.

8.5 — Weekday / Weekend

FORMULA

Weekday/Weekend = IF(WEEKDAY(A2,2)<=5, "Weekday", "Weekend")

What this does: WEEKDAY(date,2) returns 1 for Monday through 7 for Sunday; any result of 5 or below is a weekday, and anything above is a weekend. This column supports a specific analytical question: is air quality meaningfully different on weekdays, when traffic and industrial activity are highest, compared to weekends?

09 — Exploratory Data Analysis

Eleven PivotTable and PivotChart views were built across the engineered columns to surface city, seasonal, monthly, categorical, and correlational patterns in the cleaned dataset.

9.1 — Average AQI by City

Rank	City	Avg. AQI	Rank	City	Avg. AQI
1 (lowest)	Bengaluru	68.4	8	Kolkata	157.2
2	Chennai	79.1	9	Ahmedabad	161.3
3	Hyderabad	87.4	10	Jaipur	180.2
4	Pune	97.6	11	Lucknow	218.0
5	Bhopal	109.4	12	Patna	219.6
6	Mumbai	115.3	13	Kanpur	236.1
7	Chandigarh	141.4	14 (highest)	Delhi	247.9

Delhi's average AQI is roughly 3.6× Bengaluru's — the widest gap in the dataset between any two cities.

9.2 — AQI Category Distribution (all cities, 2022–2023)

Category	Record Count	% of Total
Satisfactory	3,959	38.7%
Moderate	2,760	27.0%
Poor	1,432	14.0%
Very Poor	1,423	13.9%
Good	591	5.8%
Severe	54	0.5%
Severe+	1	<0.1%

9.3 — Average AQI by Season

Season	Avg. AQI
Winter (Dec–Feb)	230.0
Post-Monsoon (Oct–Nov)	197.4
Summer (Mar–May)	151.0
Monsoon (Jun–Sep)	70.6

9.4 — Average AQI by Month

Month	Avg. AQI	Month	Avg. AQI
January	239.7	July	64.2
February	198.9	August	64.5
March	148.7	September	79.3
April	155.4	October	176.1
May	148.9	November	219.5
June	74.7	December	248.5

December recorded the highest average AQI of any month (248.5), in line with the winter seasonality pattern above.

9.5 — Average PM2.5 and PM10 by City (top 5 highest)

City	Avg. PM2.5	City	Avg. PM10
Delhi	121.1	Delhi	223.9
Kanpur	112.8	Kanpur	208.1
Lucknow	101.6	Lucknow	188.0
Patna	103.4	Patna	192.8
Jaipur	82.5	Jaipur	153.1

Delhi and Kanpur lead on both PM2.5 and PM10, which may point to shared sources such as vehicle emissions, industrial activity, and construction dust across both cities.

9.6 — Year-over-Year AQI Comparison

Year	Avg. AQI
2022	151.2
2023	151.6

AQI increased marginally in 2023 versus 2022, suggesting air quality did not meaningfully improve year-over-year across the 14-city sample.

9.7 — Weekday vs. Weekend AQI Comparison

Day Type	Avg. AQI
Weekday	150.7
Weekend	153.0

The two figures are close to identical, suggesting that weekend traffic reduction alone isn't enough to meaningfully shift AQI — a signal that continuous sources (industrial and residential emissions) may outweigh commute-driven traffic as pollution drivers in this dataset.

9.8 — Correlation Between AQI, Temperature, and Humidity

Variable Pair	Correlation (r)
AQI & Temperature	-0.58
AQI & Humidity	-0.48

Both relationships are moderate and negative: AQI tends to worsen in cooler, drier conditions, consistent with the winter-peak pattern observed in Sections 9.3 and 9.4.

10 — Dashboard Design

A single-page dashboard was built in Microsoft Excel Online, aiming to surface the most relevant air quality metrics in a clear, scannable layout without requiring the viewer to open any underlying PivotTable.

Element	Detail
KPI Cards	Average AQI (151.4), Average PM2.5 (73.3), Average PM10 (135.8), Total Records (10,220), Cities Covered (14)
Horizontal Bar Charts	City-wise AQI and PM comparisons
Column Chart	Monthly / seasonal AQI trend
Line Chart	Year-over-year AQI trend
Doughnut Chart	AQI Category distribution

11 — Key Findings & Recommendations

#	Finding	Recommendation
1	Delhi recorded the highest average AQI (247.9); Bengaluru the lowest (68.4) — a ~3.6× gap.	Prioritize pollution-control measures in Delhi and similarly high-AQI cities, targeting major emission sources.
2	"Satisfactory" was the most common AQI category (38.7%), followed by "Moderate" (27.0%).	Maintain current air quality standards while working to reduce Moderate and Poor AQI days.
3	Winter months show the highest average AQI (230.0), consistent with seasonal smog and reduced wind dispersion.	Strengthen winter pollution-control measures, including stricter emission monitoring and public advisories.
4	PM2.5 is highest in Delhi, followed by Kanpur — both dense, high-traffic industrial hubs.	Reduce fine particulate emissions through cleaner transport, industrial controls, and dust management.
5	PM10 follows the same pattern, again led by Delhi and Kanpur, pointing to shared sources.	Improve road dust management, construction-site regulation, and industrial emission compliance.
6	2023 recorded a marginally higher average AQI than 2022 — no year-over-year improvement.	Look into the causes of the year-over-year increase and consider targeted environmental interventions.
7	December had the highest average AQI of any month (248.5), reinforcing winter seasonality.	Launch seasonal pollution-control campaigns before winter and increase public awareness during peak months.
8	Bengaluru consistently had the lowest average AQI of all 14 cities across both years.	Study Bengaluru's air-quality practices and identify strategies that could transfer to more polluted cities.
9	Average AQI is nearly identical on weekdays (150.7) and weekends (153.0).	Address continuous pollution sources such as industry and residential emissions rather than traffic alone.
10	AQI is moderately negatively correlated with temperature ($r = -0.58$) and humidity ($r = -0.48$).	Incorporate weather forecasts into pollution management plans and issue early warnings during high-risk conditions.

12 — Challenges Faced

- Learning Excel dashboard creation from scratch, including KPI card layout and chart selection.
- Designing a clean, single-page dashboard layout within Excel's native visual constraints.
- Creating PivotTables and selecting appropriate chart types for each analytical question.
- Organizing eight separate worksheets for documentation, decisions, and analysis without losing traceability.
- Working around Excel Online's limitations — it does not support moving slicers across worksheets or linking text boxes dynamically to cells — which required alternative layout approaches while keeping the presentation professional.

13 — Limitations and Notes

- Missing values in PM2.5, PM10, NO2, SO2, CO, O3, Temperature, and Humidity (all under 1% per column) were retained as blanks rather than imputed. Any aggregate calculation on these columns should account for a small number of excluded rows.
- Excel Online's slicer and text-box linking limitations meant some dashboard interactivity that would be possible in desktop Excel or a dedicated BI tool (e.g. Power BI, Looker Studio) wasn't available here; documented workarounds were used in its place.
- Since the underlying pollutant readings are a synthetic dataset, the specific numeric findings above should be read as an analytical exercise rather than a real-world environmental record — the methodology, however, should carry over to production air-quality data.

14 — Conclusion

This project worked through a complete Excel-based air quality analysis workflow: baseline data profiling, missing-value and outlier assessment, feature engineering, exploratory data analysis across eleven PivotTable views, and dashboard design — corresponding to the five-stage methodology outlined in Section 2. Each decision, including the choice to retain rather than impute values under a 1% missing-data threshold, was made with transparency and data integrity in mind. The analysis points to a fairly clear picture: Delhi and Bengaluru sit at opposite ends of the pollution spectrum, winter is consistently the most polluted season, and weekday-weekend traffic differences appear to have little bearing on AQI compared to continuous industrial and residential emission sources.

Working within Excel Online's limitations, as noted in Section 12, reasonable workarounds were used to put together a structured, portfolio-ready analytical project.